Module Review: Bacterial Undecaprenyl-Phosphate Carrier Supply in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** Bacterial undecaprenyl-phosphate (C55-P) carrier supply **Source bucket queried:** KEGG ppu00550 (Peptidoglycan biosynthesis) **Module area:** lipid / cell-envelope metabolism **Date:** 2026-09-01

1. Executive summary

The generic two-step module — (1) UppS synthesis of undecaprenyl-diphosphate (C55-PP) and (2) UppP/BacA dephosphorylation to undecaprenyl-phosphate (C55-P) — is **satisfiable** in *P. putida* KT2440. Both steps are represented by reviewed (Swiss-Prot) orthologs: **uppS / PP_1595 (Q88MH6, EC 2.5.1.31)** and **uppP / bacA / PP_2862 (Q88IY7, EC 3.6.1.27)**. All KT2440 evidence is **homology-inferred** (both entries: protein existence "Inferred from homology", annotation score 3.0/5); there is no direct biochemical or genetic characterization in this strain.

Two curation-relevant nuances dominate this review:

1. **Boundary/over-collection problem.** Only 2 of the 23 candidate genes actually belong to this module. The candidate list is essentially the full peptidoglycan-biosynthesis bucket (ppu00550) plus adjacent PBP/cell-division genes. The other 21 genes are cytoplasmic Mur-ligase precursor synthesis, C55-P *consumers* (mraY, murG), or periplasmic glycan polymerases/transpeptidases — all explicitly outside the carrier-supply scope.
2. **Asymmetric redundancy.** Step 1 is **single-copy and non-redundant** (UppS is the only cis-prenyltransferase in the genome → expected essential). Step 2 is **functionally redundant**: beyond uppP (BacA family), KT2440 encodes three PAP2-family phosphatases (PP_0251, PP_0900, PP_4813) that belong to the same family (Pfam PF01569) as *E. coli*'s accessory C55-PP phosphatases PgpB/YbjG/LpxT. uppP is therefore probably individually dispensable.

Recommended module calls: Step 1 → **covered** (high confidence). Step 2 → **covered** (primary, uppP) with an attached **candidate_uncertain** note for accessory PAP2 redundancy. Module boundaries need tightening (**module_needs_revision** at the level of the bucket→module mapping, not the biology).

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2. Target-organism pathway definition

Included process (this module). Supply of the reusable C55 polyprenyl-phosphate lipid carrier for cell-envelope glycan assembly, comprising exactly two reactions:

- **C55-PP synthesis:** *trans,trans*-farnesyl-diphosphate (FPP) + 8 isopentenyl-diphosphate (IPP) → ditrans,octacis-undecaprenyl-diphosphate (C55-PP) + 8 PPi (EC 2.5.1.31; cytoplasmic-facing cis-prenyltransferase).
- **C55-P formation:** C55-PP + H_2O → C55-P + Pi (EC 3.6.1.27; integral-membrane pyrophosphatase). This activity acts on both *de novo* C55-PP (from UppS) and *recycled* C55-PP released on the periplasmic face after each round of glycan transfer.

Explicitly excluded (kept in neighboring pathways): - **Upstream:** IPP/DMAPP and FPP supply — FPP is made by IspA (PP_0528, EC 2.5.1.1/2.5.1.10). This is terpenoid-backbone/isoprenoid metabolism (KEGG ppu00900 / MEP pathway), not this module. - **Downstream / carrier consumers:** MraY (PP_1334, EC 2.7.8.13; C55-P → lipid I) and MurG (PP_1337, EC 2.4.1.227; lipid I → lipid II) *use* C55-P but perform pathway-specific sugar transfer and are outside scope. Likewise all PG polymerases/transpeptidases (ftsW, mrdB/rodA, ftsI, mrdA-I/II, mrcA/mrcB, pbpC, mtgA) and carboxypeptidases (dacA, dacB), and the cytoplasmic Mur ligases (murA–F, ddlA/B). - **Overview maps to keep separate:** KEGG global/overview maps ppu01100 (metabolic pathways), ppu01110 (biosynthesis of secondary metabolites); and the PG super-pathway maps ppu01501/ppu01502 (β-lactam resistance / peptidoglycan). Note uppP is bucketed by KEGG under **ppu00552** (not ppu00550), reflecting its carrier-recycling role.

Alternate names / database definitions. UppS = undecaprenyl pyrophosphate synthase / di-*trans*,poly-*cis*-undecaprenyl-diphosphate synthase / cis-IPPS (Pfam PF01255, "UPP synthase family"; MetaCyc UDP synthase). UppP = BacA / Upk / undecaprenyl pyrophosphate phosphatase / "bacitracin resistance protein" (Pfam PF02673, InterPro IPR003824). The carrier itself is variously "undecaprenyl phosphate", "bactoprenol phosphate", "C55-P", or "lipid carrier".

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3. Expected step model

Step	Reaction (EC)	Expected player	KT2440 status
1. Undecaprenyl-diphosphate synthesis	FPP + 8 IPP → C55-PP (2.5.1.31)	UppS cis-prenyltransferase	**covered** — uppS/PP_1595 (single-copy, non-redundant)
2. Undecaprenyl-phosphate formation	C55-PP → C55-P + Pi (3.6.1.27)	UppP/BacA + PAP2 accessory phosphatases	**covered (primary)** uppP/PP_2862; **candidate_uncertain** accessory PP_0251/PP_0900/PP_4813

No additional obligatory steps exist in the generic module. There is no expectation of a lineage-specific replacement for the *synthesis* step in a Gram-negative γ-proteobacterium such as *P. putida*; the UppS cis-prenyltransferase family is universal in bacteria.

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4. Candidate genes and evidence

High-confidence, in-scope

uppS — PP_1595 — Q88MH6 — EC 2.5.1.31 (Step 1). - **Role:** sole cis-prenyltransferase producing C55-PP. Pfam PF01255; InterPro IPR001441/IPR018520/IPR036424 ("UPP synthase family"). GO:0008834 (di-trans,poly-cis-undecaprenyl-diphosphate synthase activity), GO:0016094 (polyprenol biosynthesis), GO:0000287 (Mg2+), GO:0071555 (cell-wall organization). - **Evidence type:** ortholog assignment, reviewed Swiss-Prot; protein existence "inferred from homology" (no direct KT2440 assay). Strong homology transfer from well-characterized bacterial UppS (structural/enzymological work in *E. coli*, *S. pneumoniae*, *S. aureus*: PMIDs 24827744, 25287857, 34473495). - **Caveats:** KEGG places uppS in ppu00900 (terpenoid backbone), which is why it is not among the ppu00550 "primary" genes; this is a bucket artifact, not a functional discrepancy. Single-copy → **expected essential** (universal essentiality of UppS; validated antibiotic target). No isozyme backup exists in the genome.

uppP / bacA / upk — PP_2862 — Q88IY7 — EC 3.6.1.27 (Step 2). - **Role:** primary (BacA-family) undecaprenyl-pyrophosphate phosphatase. Pfam PF02673; InterPro IPR003824 ("UppP family"). GO:0050380 (undecaprenyl-diphosphatase activity), GO:0046677 (response to antibiotic), GO:0005886 (plasma membrane). - **Evidence type:** ortholog assignment, reviewed Swiss-Prot; "inferred from homology". The "bacitracin resistance protein" name reflects the well-established link between C55-PP phosphatase level and bacitracin resistance (bacitracin sequesters C55-PP). - **Caveats:** the single-gene annotation **under-represents the true biology of the step** — in *E. coli* BacA supplies only ~75% of membrane C55-PP phosphatase activity and a *bacA* deletion is viable (P 15778224). uppP alone is therefore probably **not essential** in KT2440.

Accessory / redundancy candidates (in-scope for Step 2, currently unannotated for it)

Gene	Acc	Family	Note
PP_0251	Q88R82	PAP2 (PF01569 / IPR000326) + haloperoxidase domain	candidate accessory C55-PP phosphatase
PP_0900	Q88PF0	PAP2 (PF01569 / IPR000326)	candidate accessory C55-PP phosphatase
PP_4813	Q88DL2	PAP2 (PF01569) fused to DedA (PF09335)	candidate; DedA fusion suggests membrane/lipid role

These share the exact family (PF01569) of *E. coli* PgpB/YbjG/LpxT, which provide the remaining ~25% of C55-PP phosphatase activity; only *bacA+ybjG+pgpB* triple inactivation is lethal (P 15778224; Manat et al. 2015, P 26560897). Transfer to *P. putida* is **family-level (moderate)**, not orthology- or function-level. Their *primary* physiological substrate (phosphatidylglycerol-phosphate, lipid A 1-phosphate, or C55-PP) is unresolved.

Out-of-scope candidates (correct annotations, wrong module)

- **C55-P consumers:** mraY/PP_1334, murG/PP_1337 (belong to lipid-linked PG precursor assembly).

- **Cytoplasmic precursor synthesis:** murA/PP_0964, murB/PP_1904, murC/PP_1338, murD/PP_1335, murE/PP_1332, murF/PP_1333, ddlA/PP_4346, ddlB/PP_1339.
- **PG polymerization/crosslinking/hydrolysis:** ftsW/PP_1336, mrdB/PP_4806, ftsI/PP_1331, mrdA-I/PP_3741, mrdA-II/PP_4807, mrcA/PP_5084, mrcB/PP_4683, pbpC/PP_0572, mtgA/PP_5107, dacA/PP_4803, dacB/PP_2098.

None of these should be attached to the carrier-supply module.

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5. Gaps, ambiguities, and likely over-annotations

- **Over-collection (primary issue).** The bucket→module mapping pulls the entire ppu00550 PG-biosynthesis set into a module whose true membership is 2 genes. This is a **module_needs_revision** at the mapping level.
- **Step-2 under-annotation.** Only uppP carries GO:0050380. The genome's PAP2 paralogs are candidate accessory phosphatases but lack the C55-PP-phosphatase annotation. This is the opposite risk to over-propagation: a real, redundant activity is likely **under-captured**.
- **PAP2 over-generalization risk.** Conversely, do **not** blanket-assign GO:0050380 to PP_0251/PP_0900/PP_4813 without phylogenetic/experimental support — PAP2 enzymes are promiscuous and many act primarily on glycerophospholipids or LPS, not C55-PP. Broad EC/GO transfer here would be a new over-propagation.
- **EC/GO breadth on PBPs.** Several out-of-scope PBPs carry broad EC 2.4.99.28 / 3.4.16.4 mappings (e.g., mrcA with both), but these are irrelevant to this module.
- **No direct KT2440 evidence.** Both in-scope genes are homology-inferred. No strain-specific essentiality dataset was retrievable in this review to confirm UppS essentiality or uppP dispensability in KT2440 specifically.

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6. Module and GO-curation recommendations

Module step calls: - **Step 1 (undecaprenyl-diphosphate synthesis):** `covered` — uppS/PP_1595, high confidence, single-copy, expected essential. - **Step 2 (undecaprenyl-phosphate formation):** `covered` via uppP/PP_2862, with an attached `candidate_uncertain` annotation recording probable accessory redundancy from PP_0251/PP_0900/PP_4813 and the resulting likelihood that uppP is individually non-essential. - **Module boundary:**

`module_needs_revision` — restrict membership to UppS + UppP/PAP2 and explicitly exclude C55-P consumers (mraY, murG) and PG polymerases. Add an explicit note that FPP supply (IspA/ppu00900) is the upstream boundary and glycan transfer is the downstream boundary.

GO / annotation actions: - Consider a curator-reviewed GO:0050380 (undecaprenyl-diphosphatase activity) annotation with an appropriate "inferred from sequence/structural similarity" evidence code for the best-supported PAP2 paralog(s), *after* orthology analysis — do not auto-propagate. - Keep uppP's KEGG bucket (ppu00552) note; the ppu00552 vs ppu00550 split reflects synthesis vs recycling and is informative, not an error. - No new GO *term* requests appear necessary — existing terms (GO:0008834, GO:0016094, GO:0050380) cover both steps. A new **module document** narrowing the reusable carrier-supply module is warranted.

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7. Genes to promote to full `fetch-gene` review

1. **uppS / PP_1595 (Q88MH6)** — confirm essentiality call and structural/active-site conservation; anchor gene of Step 1.
2. **uppP / PP_2862 (Q88IY7)** — confirm BacA-family active-site residues (Glu/Ser/Arg; P 26560897) and dispensability expectation; anchor gene of Step 2.
3. **PP_0251 (Q88R82), PP_0900 (Q88PF0), PP_4813 (Q88DL2)** — PAP2-family paralogs; promote for reciprocal-best-hit/phylogenetic assignment vs *E. coli* PgpB/YbjG/LpxT and substrate-specificity assessment before any C55-PP-phosphatase annotation.

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8. Key references

- El Ghachi M, Derbise A, Bouhss A, Mengin-Lecreulx D. *Identification of multiple genes encoding membrane proteins with undecaprenyl pyrophosphate phosphatase (UppP) activity in Escherichia coli.* J Biol Chem. 2005. P 15778224. (BacA supplies most, not all, C55-PP phosphatase activity; *bacA* viable; *bacA+ybjG+pgpB* lethal.)

- Manat G, El Ghachi M, Auger R, et al. *Membrane Topology and Biochemical Characterization of the E. coli BacA Undecaprenyl-Pyrophosphate Phosphatase.* 2015.

  P 26560897

  (BacA ≈75% of activity; PAP2 family = PgpB/YbjG/LpxT; catalytic residues.)
- Sinko W, Wang Y, Zhu W, et al. *Undecaprenyl diphosphate synthase inhibitors: antibacterial drug leads.* J Med Chem. 2014.

  P 24827744

  (UppS essential for lipid I/II and PG; drug target.)
- Danley DE, Baima ET, Mansour M, et al. *Discovery and structural characterization of an allosteric inhibitor of bacterial cis-prenyltransferase.* 2015.

  P 25287857

  ("UPPs is an essential enzyme in a key bacterial cell wall synthesis pathway.")
- Workman SD, Day J, Farha MA, et al. *Structural Insights into the Inhibition of Undecaprenyl Pyrophosphate Synthase from Gram-Positive Bacteria.* 2021.

  P 34473495
- UniProtKB (organism 160488): Q88MH6 (uppS/PP_1595), Q88IY7 (uppP/PP_2862), Q88R82/Q88PF0/Q88DL2 (PAP2 paralogs) — accessed 2026-09-01.

Evidence-strength summary

Claim	Basis	Strength for KT2440
uppS/PP_1595 makes C55-PP	UniProt ortholog + universal UppS biology	Strong (homology); direct assay absent
uppS single-copy/essential	genome prenyltransferase census + general UppS essentiality	Strong inference; not strain-verified
uppP/PP_2862 is primary C55-PP phosphatase	UniProt ortholog (BacA family)	Strong (homology)
uppP individually non-essential (redundancy)	*E. coli* genetics (P 15778224) + KT2440 PAP2 paralogs present	Moderate (family transfer)
PP_0251/0900/4813 act on C55-PP	PF01569 family match to PgpB/YbjG/LpxT	Weak–moderate; needs orthology/experiment